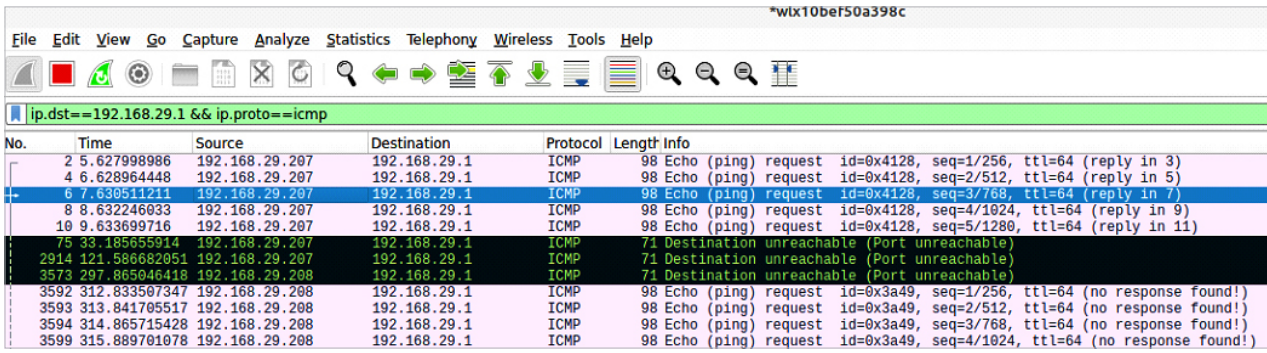

Figure 5: Output of Wireshark

This indicates that in the *POST_ROUTING* hook, the packet is dropped and not sent to wire. A log from *dmesg* command is shown in Figure 4 and describes the functionality of the module.

In this example you can also see the use of *NF_DROP* or *NF_ACCEPT* return values. The meaning of these values is self-explanatory. But there are a few more return values as well, which include:

- *NF_REPEAT:* Repeat the hook function.
- *NF_QUEUE:* Queue the packet for user space processing. To implement this in user space, we need to use the libraries *nfnetlink* and *netfilter_queue.*
- *NF_STOLEN:* Further processing of the packet and freeing memory is up to your module.

Packet mangling

Netfilter can also be used for packet mangling or modification. In the same URL *(https://github.com/SupriyoGanguly/Linux-Firewall-by-netfilter)*, you can find one more file *Mangle.c*. The corresponding makefile is *Makefile_mangle.*

In this hook, the ICMP ping packet source IP is modified just before sending out the packet. You can see the code below:

```
/* This function to be called by hook. */

static unsigned int main_hook(void *priv, struct sk_buff
*skb, const struct nf_hook_state *state)
{

     struct iphdr *ip_header = (struct iphdr *)skb_
network_header(skb);

     if (ip_header->protocol == IPPROTO_ICMP) {
     printk(KERN_INFO "Mangle icmp packet. %x\n",ip_
header->saddr);
      ip_header->saddr = 0xd01da8c0;
     }

     return NF_ACCEPT;
}
```

The output of Wireshark capture shown in Figure 5 depicts that before loading this module, the ping request to destination IP 192.168.29.1 goes from the original IP of the interface, i.e., 192.168.29.207. But after loading the module, the ping request goes from the modified IP of the interface, i.e., 192.168.29.208. However, the physical interface IP is unchanged.

Compiling the code

To compile and test the downloaded module, just use:

```
$ make

$ sudo insmod firewall
```

To remove it, use the following command:

```
$ sudo rmmod firewall
```

This article is a simple tutorial on building firewall modules using Netfilter. You can also do packet capturing by simply using the *NF_INET_PRE_ROUTING* hook number in this example. You can even use this example to simulate a man-in-the-middle attack for your devices, to test the cybersecurity. END

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